

ALLENHOUSE BUSINESS SCHOOL

Department of Bachelor of Computer Applications (BCA)

MINOR PROJECT SYNOPSIS

AgriScan AI

An AI-Based Crop Disease Detection and Advisory Web Platform

**Submitted in partial fulfilment of the requirements for the degree of
Bachelor of Computer Applications**

Submitted By

Om Awasthi (24BCA042)

Adesh Kumar (24BCA095)

Sakshi Dwivedi (24BCA139)

Madhu Devi (24BCA202)

Project Mentor: Mr. Amit Sinha

Semester: 5th Semester

Session: 2026

BONAFIDE CERTIFICATE

This is to certify that the minor project titled “AgriScan AI – An AI-Based Crop Disease Detection and Advisory Web Platform” has been carried out by Om Awasthi (24BCA042), Adesh Kumar (24BCA095), Sakshi Dwivedi (24BCA139), and Madhu Devi (24BCA202), students of BCA 5th Semester, Allenhouse Business School, under the guidance of Mr. Amit Sinha. The work presented in this synopsis is original and has not been submitted elsewhere for the award of any other degree or diploma.

Project Mentor

Mr. Amit Sinha

Assistant Professor

Department of BCA

Allenhouse Business School

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Abstract

AgriScan AI is an AI-powered web platform that detects crop diseases from images of plant leaves and provides treatment and prevention recommendations. The system uses a Convolutional Neural Network (CNN) with transfer learning trained on the PlantVillage dataset. The solution is designed to be lightweight, affordable, and easy to use for farmers through a smartphone-based web interface.

Introduction

Indian agriculture suffers significant crop losses due to late or inaccurate disease diagnosis. Farmers often depend on informal advice, which can lead to incorrect pesticide use and reduced productivity. AgriScan AI provides a fast and accessible diagnosis system that enables users to upload a crop leaf image and receive an instant disease prediction with actionable guidance.

Objectives

- Detect common crop leaf diseases using AI.
- Provide instant diagnosis through a web application.
- Suggest treatment and prevention measures.
- Build a mobile-friendly, low-cost platform.
- Allow future expansion to more crops, diseases, and regional languages.

Feasibility Study

Technical feasibility is achieved using open-source technologies such as Python, TensorFlow, Flask/FastAPI, and React.js. Economic feasibility is high because the project can be developed and deployed using free or low-cost resources. Operational feasibility is ensured through a simple user interface that requires only a smartphone and internet connection.

Methodology

The project follows the stages of data collection, preprocessing, CNN model training using transfer learning, model evaluation, backend API development, advisory mapping, frontend development, and integration testing.

Proposed System

The system consists of three layers: (1) Machine Learning Layer for disease detection, (2) Backend API Layer for processing uploaded images and returning predictions, and (3) Frontend Layer for user interaction and advisory display.

Module-wise Description

Data Acquisition and Preprocessing Module

Model Training and Evaluation Module

Inference API Module

Advisory Knowledge Base Module

Frontend/User Interface Module

Admin/Content Management Module (Future Extension)

Technology Stack

Programming Language: Python 3.x

ML Framework: TensorFlow/Keras

Model: CNN with Transfer Learning (MobileNetV2/ResNet50)

Backend: Flask/FastAPI

Frontend: React.js / HTML, CSS, JavaScript

Dataset: PlantVillage (Kaggle)

Tools: VS Code, Git, GitHub, Postman

Conclusion&References

AgriScan AI is a practical and technically feasible minor project that combines AI and web development to solve a real agricultural problem. It demonstrates machine learning, backend API integration, and responsive web application development.

References:

1. PlantVillage Dataset (Kaggle)
2. TensorFlow Documentation
3. Flask/FastAPI Documentation
4. React.js Documentation